

LOCALIZED STATIONARY PROFILES IN A SINGULAR DIFFUSION LIMIT OF THE BRUSSELTOR

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ABSTRACT. We consider the classical one-dimensional Brusselator at fixed feed parameters $A = B = 1$, with activator diffusivity $d > 0$ and inhibitor diffusivity one. Assuming the computer-assisted Core Lemma stated below, we prove that for every sufficiently small $d > 0$ the homogeneous state $(1, 1)$ possesses a nonconstant even localized stationary profile with positive concentrations. The proof rescales the joint limit $d \rightarrow 0^+$ to a fixed reversible saddle-focus core, continues one transverse intersection of its unstable manifold with the reverser fixed set, and reflects the resulting half-orbit. The continued profiles are uniformly exponentially localized and satisfy

$$\begin{aligned} \|u_d - 1\|_\infty &= \Theta(d^{1/2}), & \|v_d - 1\|_\infty &= \Theta(d), \\ W_u(d), W_v(d) &= \Theta(d^{1/4}), \end{aligned}$$

where W_u, W_v are central connected half-height widths. The result is a stationary existence theorem; it does not assert temporal stability, dynamical selection from the Turing instability, or a canard interpretation of the localized orbit.

1. INTRODUCTION

The Brusselator is a standard activator–inhibitor model for diffusion-driven pattern formation. We study the one-dimensional system

$$\begin{aligned} u_t &= du_{xx} - 2u + 1 + u^2v, \\ v_t &= v_{xx} + u - u^2v, \end{aligned} \quad x \in \mathbb{R}, \quad (1.1)$$

whose homogeneous state is $(u, v) = (1, 1)$. Our parameter is the activator diffusivity d , and the inhibitor diffusivity has been normalized to one. The question is whether a particular homoclinic geometry at the singular limit persists for all sufficiently small positive d and yields a genuine positive-concentration stationary profile of (1.1).

Localized Brusselator states are not new. Reversible 1:1 resonance gives small homoclinic solutions near an ordinary Turing point [4, 3]; spike, mesa,

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snaking, and two-dimensional localized regimes have also been studied analytically and numerically [8, 9, 6, 1, 10, 7]. On a bounded domain, computer-assisted branches of stationary and periodic Brusselator solutions and Hopf bifurcations have been proved in [2]. The singular pulses in [8] instead use a finite interval, boundary inhibitor feed, global activator feed, and simultaneously small feed parameters; their strong activator spikes and $\mathcal{O}(1)$ inhibitor scale do not give the fixed- $A = B = 1$ whole-line branch studied here. The regime here has an additional singular quantifier. If $\varepsilon = \sqrt{d}$, the Turing value for the general Brusselator at $A = 1$ is

$$B_T = (1 + \varepsilon)^2, \quad B - B_T = -2\varepsilon - \varepsilon^2 \quad \text{when } B = 1. \quad (1.2)$$

Thus $d \rightarrow 0^+$ follows a prescribed path that approaches the Turing curve. The physical critical wave number $k_T = d^{-1/4}$ diverges; equivalently, the wave number $\varepsilon k_T = d^{1/4}$ in the fast variable $y = x/\varepsilon$ tends to zero. The normal-form coefficients also become singular.

For each fixed sufficiently small $\varepsilon > 0$, Proposition 2.1 of [5] supplies local reversible-Turing homoclinics for $B < B_T$ and B close enough to B_T , in the appropriate subcriticality interval. That statement has the quantifier pattern

$$\varepsilon \text{ fixed} \implies \text{there is a neighborhood } \delta(\varepsilon) > 0 \text{ of } B_T.$$

It does not state a lower bound $\delta(\varepsilon) > 2\varepsilon + \varepsilon^2$, which would be needed to substitute the path (1.2). The contribution of this paper is to close precisely this joint-limit gap for one selected orbit and to translate the continuation into uniform physical scale laws.

Set

$$r = d^{1/4} = \sqrt{\varepsilon}. \quad (1.3)$$

An exact weighted scaling transforms the stationary equation into a smooth polynomial family whose $r = 0$ member is a fixed reversible Hamiltonian saddle-focus system. A computer-assisted calculation in a separate frozen source provides one symmetric homoclinic of that core and, crucially, a nonzero two-dimensional shooting determinant. We state this input as the Core Lemma in Section 3. Everything after that lemma—positive parameter invariant manifolds, reversible matching, full-line tails, positivity, and physical scale estimates—is proved analytically here.

The proof has five steps. Section 2 derives the exact positive- r spatial system. Section 3 states the sole model-specific imported input and explains its transversality content. Section 4 proves parameter-uniform local stable and unstable manifolds on fixed weighted spaces. Section 5 continues the core intersection and constructs a global symmetric homoclinic without using a positive- r first integral. Section 6 returns to the PDE and proves positivity, amplitudes, exponential localization, and widths. Section 7 separates these conclusions from stability, Turing selection, and canard questions.

2. EXACT CENTRAL SCALING AND THE MAIN RESULT

The stationary equation associated with the general Brusselator is

$$\begin{aligned} 0 &= du_{xx} - (B+1)u + A + u^2v, \\ 0 &= v_{xx} + Bu - u^2v. \end{aligned} \quad (2.1)$$

Put $\varepsilon = \sqrt{d}$, $p = \varepsilon u_x$, $q = v_x$, and $y = x/\varepsilon$. Then (2.1) is equivalent to

$$\begin{aligned} u_y &= p, & p_y &= -A + (B+1)u - u^2v, \\ v_y &= \varepsilon q, & q_y &= \varepsilon(-Bu + u^2v). \end{aligned} \quad (2.2)$$

Both the physical- x and fast- y systems are reversed by

$$\mathcal{R}(u, p, v, q) = (u, -p, v, -q). \quad (2.3)$$

We now fix $A = B = 1$, use (1.3), introduce $\xi = ry = x/r$, and set

$$u = 1 + r^2U, \quad p = r^3P, \quad v = 1 + r^4V, \quad q = r^3Q. \quad (2.4)$$

A direct expansion, with no remainder, gives the polynomial family

$$\begin{aligned} U' &= P, \\ P' &= -U^2 - V - 2r^2UV - r^4U^2V, \\ V' &= Q, \\ Q' &= U + r^2(U^2 + V) + 2r^4UV + r^6U^2V, \end{aligned} \quad (2.5)$$

where the prime denotes $d/d\xi$. We write its vector field as F_r . It is reversible under

$$\mathcal{R}(U, P, V, Q) = (U, -P, V, -Q), \quad \text{Fix } \mathcal{R} = \{P = Q = 0\}. \quad (2.6)$$

At $r = 0$, (2.5) reduces exactly to

$$U' = P, \quad P' = -U^2 - V, \quad V' = Q, \quad Q' = U. \quad (2.7)$$

The limiting system has primitive and Hamiltonian

$$\lambda_0 = P dU - Q dV, \quad H_0 = \frac{1}{2}(Q^2 - P^2) - \frac{1}{3}U^3 - UV, \quad (2.8)$$

with $\iota_{F_0} d\lambda_0 = dH_0$. We use (2.8) only to identify the imported core orbit; no Hamiltonian structure is asserted for $r > 0$.

The linearization $A_r = DF_r(0)$ has characteristic polynomial

$$\mu^4 - r^2\mu^2 + 1 \quad (2.9)$$

and, for $0 \leq r < \sqrt{2}$, the saddle-focus quartet

$$\pm\alpha(r) \pm i\beta(r), \quad \alpha(r) = \frac{1}{2}\sqrt{2+r^2}, \quad \beta(r) = \frac{1}{2}\sqrt{2-r^2}. \quad (2.10)$$

In particular, stable and unstable dimensions are both two and the spectral gap is uniform near $r = 0$.

For $0 < \eta < 2^{-1/2}$, define

$$X_\eta = \left\{ Z \in C^0(\mathbb{R}, \mathbb{R}^4) : \|Z\|_\eta := \sup_{\xi \in \mathbb{R}} e^{\eta|\xi|} |Z(\xi)| < \infty \right\}. \quad (2.11)$$

For an even stationary profile, define its central connected half-height widths by

$$\begin{aligned} W_u(d) &= \text{length } \mathcal{C}_0 \left\{ x : |u_d(x) - 1| \geq \frac{1}{2} |u_d(0) - 1| \right\}, \\ W_v(d) &= \text{length } \mathcal{C}_0 \left\{ x : |v_d(x) - 1| \geq \frac{1}{2} |v_d(0) - 1| \right\}, \end{aligned} \quad (2.12)$$

where $\mathcal{C}_0$ denotes the connected component containing zero.

Theorem 2.1 (Localized positive Brusselator profiles). *Assume the computer-assisted Core Lemma 3.1. There exist $r_0 > 0$, $C > 0$, and $0 < \eta < 2^{-1/2}$, together with a C^1 map*

$$[0, r_0] \longrightarrow X_\eta, \quad r \longmapsto Z_r,$$

where differentiability is with respect to $r = d^{1/4}$, such that:

- (1) Z_0 is the selected core orbit from Lemma 3.1. For every $0 < r \leq r_0$, Z_r is a nonconstant $\mathcal{R}$ -symmetric homoclinic orbit of (2.5), centered at the locally continued intersection $c_r = Z_r(0) \in \text{Fix } \mathcal{R}$, and $c_r \rightarrow c_0$ as $r \rightarrow 0^+$.
- (2) The tails are uniform:

$$|Z_r(\xi)| \leq C e^{-\eta|\xi|} \quad (0 \leq r \leq r_0, \xi \in \mathbb{R}), \quad \|Z_r - Z_0\|_\eta \longrightarrow 0 \quad \text{as } r \rightarrow 0^+. \quad (2.13)$$

- (3) If $d = r^4$, then

$$u_d(x) = 1 + r^2 U_r(x/r), \quad v_d(x) = 1 + r^4 V_r(x/r) \quad (2.14)$$

is a smooth, nonconstant, even stationary solution of (1.1), with $u_d(x) > 0$, $v_d(x) > 0$, and $(u_d(x), v_d(x)) \rightarrow (1, 1)$ as $|x| \rightarrow \infty$.

- (4) The amplitudes satisfy

$$\begin{aligned} d^{-1/2} \|u_d - 1\|_\infty &\longrightarrow \|U_0\|_\infty > 0, \\ d^{-1} \|v_d - 1\|_\infty &\longrightarrow \|V_0\|_\infty > 0. \end{aligned} \quad (d \rightarrow 0^+) \quad (2.15)$$

- (5) There are $0 < c < C_w < \infty$, independent of d , such that

$$d^{-1/2} |u_d(x) - 1| + d^{-1} |v_d(x) - 1| \leq C e^{-\eta|x|/d^{1/4}} \quad (x \in \mathbb{R}, 0 < d \leq r_0^4) \quad (2.16)$$

and

$$cd^{1/4} \leq W_u(d), W_v(d) \leq C_w d^{1/4} \quad (0 < d \leq r_0^4). \quad (2.17)$$

The threshold $d_0 = r_0^4$ is existential; it is not an explicitly certified diffusion interval.

3. THE COMPUTER-ASSISTED CORE INPUT

We isolate the sole model-specific imported statement. This formulation is deliberately smaller than the return–exit and coding results available in the source: those additional conclusions are not used here.

Lemma 3.1 (Computer-assisted Core Lemma). *The core system (2.7) has a selected nonconstant $\mathcal{R}$ -symmetric homoclinic orbit Γ_0 to the origin. Its center $c_0 = \Gamma_0(0)$ lies in $\text{Fix } \mathcal{R}$ and satisfies*

$$\begin{aligned} U_0(0) &\in [4.8785234574459304, 4.8785234988116768], \\ V_0(0) &\in [-7.9333304994224827, -7.933330385013492]. \end{aligned} \quad (3.1)$$

Moreover,

$$W_0^u(0) \cap \text{Fix } \mathcal{R} \quad \text{at } c_0. \quad (3.2)$$

More precisely, for the source phase ϕ and finite flight time T , the normal shooting map

$$\mathcal{M}_0(\phi, T) = (P, Q)(\Phi_0^T(S_0(\phi)))$$

has a locally unique zero (ϕ_0, T_0) in the reported shooting box and

$$\det D_{(\phi, T)} \mathcal{M}_0(\phi_0, T_0) \in [149.56393055300413, 149.56404227745782]. \quad (3.3)$$

The corresponding hit is the first positive nontrivial hit of $\text{Fix } \mathcal{R}$.

The lemma is imported from Proposition 8.6 together with the frozen certificate and validation README described in Appendix A. The manuscript's equations (8.36)–(8.40) give the phase, flight time, determinant, tangent rows, and first-hit identification. The center enclosures, local uniqueness in the shooting box, and detailed sign checks excluding an earlier hit are recorded in the certificate and README. The determinant has the following direct geometric meaning. Let

$$G(\phi, T) = \Phi_0^T(S_0(\phi)), \quad N(U, P, V, Q) = (P, Q).$$

Then $\mathcal{M}_0 = N \circ G$, $\ker DN = T \text{Fix } \mathcal{R}$, and the image of DG is the tangent space of the flowed two-dimensional unstable patch. Invertibility of $D(N \circ G)$ implies that this tangent space intersects $T \text{Fix } \mathcal{R}$ trivially, which is exactly (3.2). This is transversality to the reverser fixed set, not ambient transversality of W^u and W^s in four dimensions.

Remark 3.2. Lemma 3.1 is a computer-assisted imported input, not a floating point observation. Its frozen source records outward-rounded flow and first variation bounds, a two-dimensional Krawczyk inclusion, a rigorous local unstable graph, and first-hit sign checks. At the date of this manuscript, the immutable source revision and hashes are recorded, but the complete evidence closure is not anonymously accessible at a stable public archive. Thus Theorem 2.1 is mathematically conditional on the precisely stated Core Lemma, and a publication release must also make that lemma's evidence package externally auditable.

4. UNIFORM LOCAL INVARIANT MANIFOLDS

We prove the parameter-uniform local result needed both for continuation and for full-line tails. Write

$$F_r(Z) = A_r Z + N_r(Z), \quad N_r(0) = DN_r(0) = 0. \quad (4.1)$$

Although the physical problem has $r \geq 0$, the polynomial family extends to negative r , which permits an ordinary C^1 parameter argument at zero.

Lemma 4.1 (Uniform local manifolds). *After decreasing $r_0 > 0$, the origin of F_r has two-dimensional local stable and unstable manifolds $W_{\text{loc}}^s(r)$ and $W_{\text{loc}}^u(r)$ for $|r| \leq r_0$. They admit graph and half-orbit parameterizations over fixed two-dimensional spaces which are C^1 in the graph parameter and in r . The half-orbits and their first r -derivatives obey exponential estimates with constants independent of r . A compact patch may be flowed for every bounded time for which its $r = 0$ flowout lies in a compact subset of the flow domain, and the continued flowout retains this C^1 dependence.*

Proof. Choose $r_0 < 1$. By (2.10), the stable and unstable Riesz projections Π_r^s, Π_r^u vary smoothly. Fix

$$0 < \eta < \eta_1 < a_0 < \inf_{|r| \leq r_0} \alpha(r). \quad (4.2)$$

After decreasing r_0 , the restrictions

$$J_r^u = \Pi_r^u|_{E_0^u} : E_0^u \rightarrow E_r^u, \quad J_r^s = \Pi_r^s|_{E_0^s} : E_0^s \rightarrow E_r^s \quad (4.3)$$

are smooth isomorphisms, and their inverses are uniformly bounded. Spectral calculus in finite dimension and the Duhamel formula give K_j such that, for $j = 0, 1$,

$$\begin{aligned} \left\| \partial_r^j (e^{A_r t} \Pi_r^s) \right\| &\leq K_j (1+t)^j e^{-a_0 t} \quad (t \geq 0), \\ \left\| \partial_r^j (e^{A_r t} \Pi_r^u) \right\| &\leq K_j (1+|t|)^j e^{a_0 t} \quad (t \leq 0). \end{aligned} \quad (4.4)$$

For the unstable manifold use the fixed space

$$X_{\eta_1}^- = \left\{ Z \in C^0((-\infty, 0], \mathbb{R}^4) : \|Z\|_{\eta_1, -} = \sup_{t \leq 0} e^{-\eta_1 t} |Z(t)| < \infty \right\}. \quad (4.5)$$

Apply an $\mathcal{R}$ -invariant radial cutoff at radius 2δ to N_r , equal to N_r on the ball of radius δ , and call the result $\tilde{N}_r$. Its uniform Lipschitz constant L_δ tends to zero with δ . For $b \in E_0^u$ define

$$\begin{aligned} (\mathcal{T}_{r,b} Z)(t) &= e^{A_r t} J_r^u b + \int_0^t e^{A_r(t-s)} \Pi_r^u \tilde{N}_r(Z(s)) ds \\ &\quad + \int_{-\infty}^t e^{A_r(t-s)} \Pi_r^s \tilde{N}_r(Z(s)) ds. \end{aligned} \quad (4.6)$$

With

$$D_0 = (a_0 - \eta_1)^{-1} + (a_0 + \eta_1)^{-1}, \quad K = \max\{K_0, 1\},$$

the estimates above imply

$$\begin{aligned} \|\mathcal{T}_{r,b}Z\|_{\eta_1,-} &\leq K \|J_r^u\| |b| + KL_\delta D_0 \delta, \\ \|\mathcal{T}_{r,b}Z - \mathcal{T}_{r,b}\tilde{Z}\|_{\eta_1,-} &\leq KL_\delta D_0 \|Z - \tilde{Z}\|_{\eta_1,-}. \end{aligned} \quad (4.7)$$

Choose δ so that $KL_\delta D_0 \leq 1/2$, and then choose $b_* > 0$ so that

$$K \sup_{|r| \leq r_0} \|J_r^u\| b_* \leq \delta/2.$$

Thus (4.6) is a uniform strict contraction of one fixed closed ball for all $|r| \leq r_0$ and $|b| \leq b_*$. Denote its fixed point by $z^u(r, b)$.

The strict gaps in (4.2) also control the differentiated kernels: the factor $(1 + |t - s|)e^{-a_0|t-s|}$ remains integrable in the η_1 -weighted norm, and $\partial_r \tilde{N}_r(Z) = \mathcal{O}(|Z|^2)$ uniformly in the cutoff ball. The parameterized contraction theorem gives

$$(r, b) \mapsto z^u(r, b) \in X_{\eta_1}^- \quad \text{of class } C^1. \quad (4.8)$$

The endpoint

$$G^u(r, b) = z^u(r, b)(0) \quad (4.9)$$

parameterizes $W_{\text{loc}}^u(r)$. The fixed point lies in the region where the cutoff agrees with the original field. The stable construction is the forward-time analogue, and reversibility plus local uniqueness gives

$$W_{\text{loc}}^s(r) = \mathcal{R}W_{\text{loc}}^u(r). \quad (4.10)$$

The final assertion follows from the common-flow-domain theorem and smooth finite-time dependence for ODEs on compact subsets of the flow domain. $\square$

5. REVERSIBLE MATCHING AND THE FULL HOMOCLINIC

Let $S_0(\phi)$ be the source curve used in Lemma 3.1, with matching map $\mathcal{M}_0$. Choose a compact phase interval I containing ϕ_0 in its interior. Since $S_0(I)$ is a compact curve in $W_0^u(0)$, after shrinking I there is a common $\tau > 0$ for which

$$\hat{S}_0(\phi) = \Phi_0^{-\tau}(S_0(\phi)) = G^u(0, b_0(\phi)), \quad |b_0(\phi)| < b_*, \quad (5.1)$$

for a C^1 map $b_0 : I \rightarrow E_0^u$. Define the genuine parameter-dependent unstable source curve by

$$\hat{S}(r, \phi) = G^u(r, b_0(\phi)), \quad S(r, \phi) = \Phi_r^\tau(\hat{S}(r, \phi)). \quad (5.2)$$

Then $S(r, \phi) \in W_r^u(0)$, $S(0, \phi) = S_0(\phi)$, and the family is C^1 .

Choose a compact time interval $J \Subset (0, \infty)$ containing T_0 . After shrinking I, J , all base trajectories $\Phi_0^T(S_0(\phi)), (\phi, T) \in I \times J$, lie in a common compact flow tube. Therefore

$$\mathcal{M}(r, \phi, T) = (P, Q)(\Phi_r^T(S(r, \phi))) \quad (5.3)$$

is C^1 near $(0, \phi_0, T_0)$. Lemma 3.1 and the finite-dimensional implicit-function theorem give C^1 functions $\phi(r), T(r)$ with

$$\mathcal{M}(r, \phi(r), T(r)) = 0, \quad \phi(0) = \phi_0, \quad T(0) = T_0. \quad (5.4)$$

After decreasing r_0 , continuity of the determinant, together with the base enclosure, gives, for example,

$$\left| \det D_{(\phi, T)} \mathcal{M}(r, \phi(r), T(r)) \right| \geq 70 \quad (|r| \leq r_0). \quad (5.5)$$

This numerical constant is not a certified positive- r interval statement; it is a convenient consequence of continuity on a sufficiently small existential neighborhood.

Put

$$c_r = \Phi_r^{T(r)}(S(r, \phi(r))). \quad (5.6)$$

Then $c_r \in \text{Fix } \mathcal{R}$, $c_r \in W_r^u(0)$, and $c_r \neq 0$ for small $|r|$. The negative half-orbit exists for all negative time and converges to zero. Define the positive half by reflection:

$$Z_r(\xi) = \begin{cases} \Phi_r^\xi(c_r), & \xi \leq 0, \\ \mathcal{R}\Phi_r^{-\xi}(c_r), & \xi \geq 0. \end{cases} \quad (5.7)$$

Because $c_r \in \text{Fix } \mathcal{R}$ and $D\mathcal{R}F_r = -F_r \circ \mathcal{R}$, the values and first derivatives agree at zero. ODE uniqueness makes (5.7) a smooth, nonconstant, symmetric global homoclinic. Notice that forward global existence was constructed by reflection rather than inferred from a nonexistent positive- r energy integral.

We next record the full-line weighted dependence. The entry-to-center time $\sigma(r) = \tau + T(r)$ is bounded. Choose a fixed $T_* > \sup_{|r| \leq r_0} \sigma(r)$ large enough that

$$\begin{aligned} a_r &:= z^u(r, b_0(\phi(r)))(-(T_* - \sigma(r))) \\ &= Z_r(-T_*) \end{aligned} \quad (5.8)$$

lies strictly in the uniform graph disk. Formula (5.8) and (4.8) show directly that $r \mapsto a_r$ is C^1 ; no full-line Banach-space regularity has been used. Set

$$b_r = (J_r^u)^{-1} \Pi_r^u a_r. \quad (5.9)$$

Then b_r is C^1 and $|b_r| < b_*$. Evaluating (4.6) at $t = 0$ gives

$$\Pi_r^u G^u(r, b) = J_r^u b. \quad (5.10)$$

Since a_r lies on the local unstable graph, (5.9)–(5.10) and uniqueness of the graph parameter imply $a_r = G^u(r, b_r)$. Orbit uniqueness now gives the fixed-shift representation

$$Z_r(\xi) = z^u(r, b_r)(\xi + T_*) \quad (\xi \leq -T_*). \quad (5.11)$$

Equations (4.8), (5.11), and $\eta < \eta_1$ yield

$$\sup_{|r| \leq r_0} \sup_{\xi \leq -T_*} e^{-\eta\xi} (|Z_r(\xi)| + |\partial_r Z_r(\xi)|) < \infty. \quad (5.12)$$

Common finite-time dependence controls $[-T_*, 0]$, and reflection controls the positive half-line. Hence

$$r \mapsto Z_r \in X_\eta \quad \text{is } C^1, \quad \sup_{|r| \leq r_0} \sup_{\xi \in \mathbb{R}} e^{\eta|\xi|} (|Z_r(\xi)| + |\partial_r Z_r(\xi)|) < \infty. \quad (5.13)$$

This proves the first two assertions of Theorem 2.1.

6. RETURN TO THE PDE: POSITIVITY AND SCALE LAWS

The inverse change of variables (2.14) is exact for every $r > 0$, so Z_r gives a stationary solution of (1.1). Symmetry of Z_r gives

$$U_r(-\xi) = U_r(\xi), \quad V_r(-\xi) = V_r(\xi),$$

and hence the physical profiles are even.

From (5.13), there is $M > 0$ such that

$$\sup_{0 \leq r \leq r_0} (\|U_r\|_\infty + \|V_r\|_\infty) \leq M. \quad (6.1)$$

Decrease r_0 so that $r_0^2 M < 1/2$ and $r_0^4 M < 1/2$. Then

$$u_d(x) \geq \frac{1}{2}, \quad v_d(x) \geq \frac{1}{2} \quad (6.2)$$

for every $x \in \mathbb{R}$ and $0 < r \leq r_0$. Exponential convergence to the homogeneous state follows directly from (2.13).

Uniform convergence $Z_r \rightarrow Z_0$ and the exact identities

$$d^{-1/2} \|u_d - 1\|_\infty = \|U_r\|_\infty, \quad d^{-1} \|v_d - 1\|_\infty = \|V_r\|_\infty \quad (6.3)$$

prove (2.15). Both limits are nonzero by (3.1). Replacing ξ by x/r in (2.13), and using $r = d^{1/4}$, proves (2.16).

It remains to prove the width statement for the explicit observable (2.12). By (3.1) and uniform convergence there is $m > 0$ with

$$|U_r(0)| \geq m, \quad |V_r(0)| \geq m \quad (0 \leq r \leq r_0). \quad (6.4)$$

The weighted bound on $P_r = U'_r$ and $Q_r = V'_r$ gives a uniform derivative bound $M_1 \geq 1$. With $\ell = m/(4M_1)$, the mean-value theorem gives

$$|U_r(\xi)| \geq \frac{1}{2}|U_r(0)|, \quad |V_r(\xi)| \geq \frac{1}{2}|V_r(0)| \quad (|\xi| \leq \ell). \quad (6.5)$$

Choose $L > 0$, independent of r , so that $Ce^{-\eta L} < m/2$. The two scaled half-height inequalities fail for $|\xi| \geq L$, while (6.5) shows that their central components contain $[-\ell, \ell]$. Their lengths in ξ lie between 2ℓ and $2L$. Multiplication by $x = r\xi$ proves (2.17), and completes the proof of Theorem 2.1.

7. SCOPE AND RELATION TO TURING PATTERNS AND CANARDS

Theorem 2.1 proves the existence of a spatially localized stationary solution. It also determines how its amplitude and one precise width observable scale along the singular path. These are physical-variable consequences of the homoclinic continuation, but they do not by themselves describe what the time-dependent PDE selects from generic initial data.

Three nearby questions require additional arguments.

First, temporal stability is a spectral problem for the linearization of (1.1) about (u_d, v_d) . The instability theorem in [3] concerns specified branches in a fixed-diffusion reversible-Hopf setting; its hypotheses and singular uniformity would have to be checked before either stability or instability could be transferred to the branch constructed here.

Second, (1.2) locates the parameter path relative to the Turing curve, but the present proof starts from a selected RFSN-II core and continues its spatial homoclinic. It does not prove that a branch selected by time evolution or numerical continuation from the Turing bifurcation is the same global branch for every small d .

Third, spatially periodic canards organize important Brusselator dynamics in the same singular regime [5]. The local oscillations of the core used here arise from its saddle-focus eigenvalues, whereas a canard designation requires a verified orbit segment that tracks attracting and repelling slow manifolds through the relevant charts. No such identification is needed for Theorem 2.1, and none is claimed.

These distinctions leave a useful hierarchy. The present theorem supplies a rigorously organized stationary branch, scale laws, and a canonical profile on which later numerical and spectral studies can be based. Explicit positive- d validation, temporal stability, Turing-branch identification, and canard geometry are subsequent problems rather than omitted consequences of the theorem proved here.

APPENDIX A. FROZEN EVIDENCE PROVENANCE

The Core Lemma is imported from H. Lu, *First returns, singular exits, and action finite parts near a reversible Hamiltonian saddle-focus*, at the frozen revision of <https://github.com/h-lu/reversible-rfsn-ii-waves>,

d54add098545063d5efe8f1d6f062d4cfc116a0d.

The cited manuscript location is Proposition 8.6, specifically its verification of the transverse homoclinic tube. Equations (8.36)–(8.40) record the phase, flight time, determinant, tangent rows, and first-hit identification, including

$$\begin{aligned}\phi_0 &\in [5.8615055856447817, 5.8615055856450482], \\ T_0 &\in [9.6374420678958099, 9.6374420678971511].\end{aligned}\tag{A.1}$$

The endpoint enclosures (3.1), local uniqueness in the shooting box, and detailed early-hit sign exclusions are recorded in the frozen certificate and its README rather than in those displayed manuscript equations. The frozen SHA-256 identities are as follows:

```
source manuscript
0baf6335aad72d5893479d8876d2613671ecb8ac2ccd73664405dea4381e6a20
validation README
34f3e15f475d22be939020bde2b8480ae33ba60a9c043f577088ab5d70253610
certificate JSON
ed0f9f58f8ba5f1d5c36dc7c3a72bb725599c4172a3cd610d890b88699fecfbd
interval source
655cfe16e2cdb24185f11cbb314e7c0b2f029705d4588fdbcba5a9f0993298f3.
```

The interval calculation used outward-rounded CAPD/FILIB flow and first-variation enclosures, a two-dimensional Krawczyk inclusion, and interval sign checks for the first symmetry hit. Its rigorous local unstable graph is

a required dependency of the shooting certificate, so the certificate JSON alone is not a complete evidence archive.

The source URL and historical hashes preserve provenance. A submission release must additionally provide anonymous stable access either to the frozen source and its complete evidence closure or to an equivalent archived bundle with replay metadata. This packaging obligation affects external auditability, not the analytic implication from Lemma 3.1 to Theorem 2.1.

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